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Studierichting: Informatica
Oefeningenreeks 7C nr. 1.4

$$f : \mathbb{R}^2 \rightarrow \mathbb{R} : (x, y) \rightarrow x^4 y^3$$

$$\frac{\partial f}{\partial x} = 4x^3 y^3$$

$$\frac{\partial f}{\partial y} = 3y^2 x^4$$

$$\Rightarrow Df(x, y) = \begin{pmatrix} 4x^3 y^3 & 3x^4 y^2 \end{pmatrix}$$

$$\frac{\partial^2 f}{\partial x^2} = 12x^2 y^3$$

$$\frac{\partial^2 f}{\partial x \partial y} = 12x^3 y^2$$

$$\frac{\partial^2 f}{\partial y^2} = 6y x^4$$

$$\frac{\partial^2 f}{\partial y \partial x} = 12x^3 y^2$$

$$\Rightarrow D^2 f(x, y) = \begin{pmatrix} 12y^3 x^2 & 12y^2 x^3 \\ 12x^3 y^2 & 6x^4 y \end{pmatrix}$$

References